Digital Electronics Lab (CPE 462)

Project

Due Date (30/5/2026)

Accumulative Memory-Row Counter Using 6T SRAM and Synchronous D Flip-Flops (ABET 06)

Project Description

In this project, students will design a digital system that integrates a cyclic counter with a small SRAM array and an accumulator. The system repeatedly scans memory rows and accumulates their stored values over time.

Concept Overview

A 2-bit synchronous counter will continuously cycle through the sequence: 0→1→2→3→0..

Each counter value corresponds to a memory row address inside a 3x2 SRAM array, built using 6T SRAM cells. Start your project by writing data to the SRAM as: the word in row 1 is 11, the word in row 2 is 10, and the word in row 3 is 01.

At every clock cycle:

- The counter selects one memory row.
- The stored data in that row is read.
- The read value is latched using D flip-flops.
- The latched values are then added together using a half adder.
- The system continuously sums the memory contents in a circular manner.
- According to the saved data in the SRAM memory the output will be 10 then 01 then 01.

Key Requirements:

- Design all the blocks in the design at the schematic level and verify **their** functionality.
- Design the top-level circuit at the schematic level and demonstrate the correct functionality of the system.
- Design all the blocks in the design at the layout level and verify **their** functionality.
- Design the top-level circuit at the layout level and demonstrate the correct functionality of the system at least for 5 different cases
- Performance analysis (the propagation delay, SRAM Read Delay Analysis, Maximum Clock Frequency for counter)
- A comprehensive report explaining the design choices, implementation details, the screenshots of the blocks and the top-level design, screenshots of the output waveforms for **all** the buildings blocks and performance analysis

Evaluation Criteria:

- Correctness and efficiency of the circuit.
- Quality and clarity of the schematic and layout designs and simulation results.

Submission Rules

- Students can work in teams of 3
- Cheating will result in automatic Zero.
- Late submission is not allowed.